

```
mysql> use cg_mysql;
Database changed
```

```
mysql> CREATE TABLE Students (student_id INT PRIMARY KEY, name VARCHAR(100), age
INT);
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> CREATE TABLE Courses (course_id INT PRIMARY KEY, course_name VARCHAR(100));
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> CREATE TABLE Enrollments (enrollment_id INT PRIMARY KEY, student_id INT,
course_id INT);
Query OK, 0 rows affected (0.04 sec)
```

```
mysql>
mysql> INSERT INTO Students VALUES
-> (1, 'Arjun', 20), (2, 'Bhavana', 21), (3, 'Charan', 22), (4, 'Divya', 20),
-> (5, 'Eswar', 23), (6, 'Farhan', 21), (7, 'Gowri', 22);
Query OK, 7 rows affected (0.01 sec)
Records: 7 Duplicates: 0 Warnings: 0
```

```
mysql>
mysql> INSERT INTO Courses VALUES
-> (101, 'Mathematics'), (102, 'Physics'), (103, 'Biology'), (104,
'Chemistry'), (105, 'Computer Science');
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

```
mysql>
mysql> INSERT INTO Enrollments VALUES
-> (1, 1, 101), (2, 2, 101), (3, 2, 102), (4, 3, 103),
-> (5, 4, 101), (6, 5, 104), (7, 1, 102), (8, 6, 105);
Query OK, 8 rows affected (0.06 sec)
Records: 8 Duplicates: 0 Warnings: 0
```

```
mysql> select * from students;
```

student_id	name	age
1	Arjun	20
2	Bhavana	21
3	Charan	22
4	Divya	20
5	Eswar	23
6	Farhan	21
7	Gowri	22

```
7 rows in set (0.00 sec)
```

```
mysql> select * from courses;
```

course_id	course_name
101	Mathematics
102	Physics
103	Biology
104	Chemistry
105	Computer Science

5 rows in set (0.00 sec)

```
mysql> select * from enrollments;
```

enrollment_id	student_id	course_id
1	1	101
2	2	101
3	2	102
4	3	103
5	4	101
6	5	104
7	1	102
8	6	105

8 rows in set (0.00 sec)

```
mysql> select s.name, c.course_name from enrollments e join students s on
e.student_id = s.student_id join courses c on e.course_id = c.course_id;
```

name	course_name
Arjun	Mathematics
Bhavana	Mathematics
Bhavana	Physics
Charan	Biology
Divya	Mathematics
Eswar	Chemistry
Arjun	Physics
Farhan	Computer Science

8 rows in set (0.00 sec)

```
mysql> select c.course_name, s.name from courses c join enrollments e on
c.course_id = e.course_id join students s on e.student_id = s.student_id;
```

course_name	name
Mathematics	Arjun
Mathematics	Bhavana
Physics	Bhavana

Biology	Charan
Mathematics	Divya
Chemistry	Eswar
Physics	Arjun
Computer Science	Farhan

8 rows in set (0.00 sec)

```
mysql> select s.name from students s join enrollments e on s.student_id =
e.student_id join courses c on e.course_id = c.course_id where c.course_name =
'Physics';
```

name
Bhavana
Arjun

2 rows in set (0.05 sec)

```
mysql> select s.name, c.course_name from students s left join enrollments e on
s.student_id = e.student_id left join courses c on e.course_id = c.course_id;
```

name	course_name
Arjun	Physics
Arjun	Mathematics
Bhavana	Physics
Bhavana	Mathematics
Charan	Biology
Divya	Mathematics
Eswar	Chemistry
Farhan	Computer Science
Gowri	NULL

9 rows in set (0.00 sec)

```
mysql> select c.course_name, s.name from courses c left join enrollments e on
c.course_id = e.course_id left join students s on e.student_id = s.student_id;
```

course_name	name
Mathematics	Divya
Mathematics	Bhavana
Mathematics	Arjun
Physics	Arjun
Physics	Bhavana
Biology	Charan
Chemistry	Eswar
Computer Science	Farhan

8 rows in set (0.00 sec)

```
mysql> select s.name, c.course_name from students s left join enrollments e on  
s.student_id = e.student_id right join courses c on e.course_id = c.course_id;
```

name	course_name
Arjun	Mathematics
Bhavana	Mathematics
Divya	Mathematics
Bhavana	Physics
Arjun	Physics
Charan	Biology
Eswar	Chemistry
Farhan	Computer Science

8 rows in set (0.00 sec)

```
mysql> select s.name, c.course_name from students s left join enrollments e on  
s.student_id = e.student_id left join courses c on e.course_id = c.course_id
```

-> union

```
-> select s.name, c.course_name from students s right join enrollments e on  
s.student_id = e.student_id right join courses c on e.course_id = c.course_id;
```

name	course_name
Arjun	Physics
Arjun	Mathematics
Bhavana	Physics
Bhavana	Mathematics
Charan	Biology
Divya	Mathematics
Eswar	Chemistry
Farhan	Computer Science
Gowri	NULL

9 rows in set (0.00 sec)

```
mysql> select s1.name as student1, s2.name as student2, s1.age from students s1  
join students s2 on s1.age = s2.age and s1.student_id < s2.student_id;
```

student1	student2	age
Arjun	Divya	20
Bhavana	Farhan	21
Charan	Gowri	22

3 rows in set (0.03 sec)

```
mysql> select c.course_name, count(e.student_id) as total_students from courses c
```

```
left join enrollments e on c.course_id = e.course_id group by c.course_name;
```

course_name	total_students
Mathematics	3
Physics	2
Biology	1
Chemistry	1
Computer Science	1

5 rows in set (0.00 sec)

```
mysql> select s.name, count(e.course_id) as total_courses from students s left join enrollments e on s.student_id = e.student_id group by s.name;
```

name	total_courses
Arjun	2
Bhavana	2
Charan	1
Divya	1
Eswar	1
Farhan	1
Gowri	0

7 rows in set (0.00 sec)

```
mysql> select s.name, c.course_name from students s left join enrollments e on s.student_id = e.student_id left join courses c on e.course_id = c.course_id order by s.name;
```

name	course_name
Arjun	Physics
Arjun	Mathematics
Bhavana	Physics
Bhavana	Mathematics
Charan	Biology
Divya	Mathematics
Eswar	Chemistry
Farhan	Computer Science
Gowri	NULL

9 rows in set (0.00 sec)

```
mysql> select s.name, c.course_name from enrollments e join students s on e.student_id = s.student_id join courses c on e.course_id = c.course_id order by c.course_name;
```

name	course_name
------	-------------

Charan	Biology
Eswar	Chemistry
Farhan	Computer Science
Arjun	Mathematics
Bhavana	Mathematics
Divya	Mathematics
Bhavana	Physics
Arjun	Physics

8 rows in set (0.00 sec)

```
mysql> select c.course_name, count(e.student_id) as total_students from courses c
join enrollments e on c.course_id = e.course_id group by c.course_name having
total_students > 1;
```

course_name	total_students
Mathematics	3
Physics	2

2 rows in set (0.00 sec)

```
mysql> select s.name, count(e.course_id) as total_courses from students s join
enrollments e on s.student_id = e.student_id group by s.name having total_courses >
1;
```

name	total_courses
Arjun	2
Bhavana	2

2 rows in set (0.00 sec)

```
mysql> select s.name, count(e.course_id) as total_enrollments from students s left
join enrollments e on s.student_id = e.student_id group by s.name order by
total_enrollments desc;
```

name	total_enrollments
Arjun	2
Bhavana	2
Charan	1
Divya	1
Eswar	1
Farhan	1
Gowri	0

7 rows in set (0.00 sec)

```
mysql> notee
```